Exploring Optimization Strategies for Strongly Convex Ridge Regression under Noise and Conditioning

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GitHub Codebase: https://github.com/royak747/ridge-optimization-lab

1. Introduction

1.1 Inspiration

Linear regression is one of the most widely used methods in supervised learning, providing a simple yet powerful framework for modeling relationships between variables. Given a dataset with feature matrix $X \epsilon \mathbb{R}^{n \times d}$ and response vector $y \epsilon \mathbb{R}^{n}$, the goal is to estimate parameters $w \epsilon \mathbb{R}^{d}$ that minimize prediction error, typically measured using the Mean Squared Error (MSE) objective.

Although this objective is convex and admits efficient optimization methods, its practical performance depends heavily on the conditioning of the data. A key drawback of linear regression arises when the feature matrix X exhibits multicollinearity, meaning that some features are highly correlated. In this case, the matrix $X^T X$, which determines the curvature of the loss function, becomes ill-conditioned. From an optimization perspective, the condition number of $X^T X$ governs the convergence rate of iterative methods such as Gradient Descent. When the condition number is large, the objective function becomes highly elongated, causing inefficient or significantly slower convergence. Additionally, in the presence of noisy or imperfect data, ill-conditioning can amplify instability in the solution. As many real-world datasets are both ill-conditioned and noisy, these challenges motivate the use of regularization techniques such as ridge regression, which improve conditioning and lead to more stable and efficient optimization.

Ridge regression addresses these challenges by introducing an L_2 regularization term that improves conditioning and guarantees strong convexity. From an optimization perspective, ridge regression modifies the curvature of the objective by replacing $X^T X$ with $X^T X + \lambda I$. Adding λI ensures that the objective is strongly convex, even when $X^T X$ is singular. Larger values of λ improve conditioning but may introduce bias, while smaller values retain data accuracy but may result in slower convergence. This makes ridge regression a useful framework for systematically studying how optimization algorithms behave under controlled levels of conditioning.

The goal of this work is to systematically study how conditioning and noise affect the convergence behavior of optimization algorithms. We focus on Gradient Descent (GD) and Stochastic Gradient Descent (SGD), comparing their empirical behavior with theoretical predictions. Our workflow begins with controlled synthetic experiments and concludes with a validation experiment on a real-world dataset.

1.2 Related Works

Recent work has focused on mitigating the effect of ill-conditioning on gradient-based optimization. In particular, scaled or preconditioned variants of gradient descent have been proposed to reduce the dependence of convergence rates on the condition number (Tong, 2022; Li & Huang, 2026). These methods modify the update direction to improve the geometry of the optimization problem, enabling faster convergence even in ill-conditioned settings, ultimately aiming to modify the optimization algorithm to overcome ill-conditioning. Our work takes a complementary perspective where, instead of designing new algorithms, we systematically study how conditioning, noise, and regularization affect the behavior of standard gradient-based methods. In particular, we construct synthetic datasets with controlled condition numbers and noise levels to isolate these effects and empirically evaluate convergence properties across Gradient Descent, ridge Gradient Descent, and Stochastic Gradient Descent.

2. Problem Formulation

We consider the ridge regression objective $f(w) = \frac{1}{n}||Xw - y||^2 + \lambda||w||^2$ where $\lambda > 0$ is the regularization parameter. The Hessian of this objective is given by $\nabla^2 f(w) = \frac{2}{n}X^T X + 2\lambda I$ which ensures that the function is strongly convex.

Let μ and L denote the smallest and largest eigenvalues of the Hessian, respectively. The condition number is defined as $\kappa = \frac{L}{\mu}$.

For Gradient Descent with an appropriate step size, the convergence rate satisfies $||w_k - w^*|| \leq Cp^k$, where p is the contraction factor, highlighting the critical role of conditioning in determining optimization efficiency.

3. Experimental Workflow

3.1. Generating Synthetic Datasets

We begin by constructing synthetic datasets with explicitly controlled conditioning and noise levels. For a given number of samples n and features d, the design matrix $X \in \mathbb{R}^{n \times d}$ is generated by singular value decomposition in the form $X = U\Sigma V^T$, where $U \in \mathbb{R}^{n \times d}$ and $V \in \mathbb{R}^{d \times d}$ are orthonormal matrices obtained via QR decomposition of random Gaussian matrices. The diagonal matrix Σ contains singular values that decay geometrically from 1 to $\frac{1}{\kappa}$, where κ is the prescribed condition number of X. This allows for precise control over the conditioning of the quadratic objective through $X^T X$. The response vector is generated according to the linear model $y = X\beta^* + \varepsilon$ where $\beta^* \sim N(0, I_d)$ and $\varepsilon \sim N(0, \sigma^2 I_n)$. By varying the conditioning number and the noise level, we isolate their effects on optimization dynamics.

3.2. Standard Gradient Descent (OLS)

We first analyze standard Gradient Descent applied to the ordinary least squares (OLS) objective $min_w \frac{1}{n}||Xw - y||^2$. This serves as a baseline to study how conditioning alone affects convergence. In particular, we track the objective value $f(w_k)$ over iterations and the distance to the empirical optimum given by $||w_k - w^*||$. This stage will highlight how ill-conditioned problems exhibit slow linear convergence governed by the condition number.

3.3. Regularized Optimization: Ridge Regression

Afterwards, we introduce L_2 regularization and consider the ridge objective $min_w \frac{1}{n}||Xw - y||^2 + \lambda||w||^2$, which we optimize using ridge GD and ridge mini-batch SGD. For each method, we compute the empirical optimum via the closed-form ridge solution $w^* = (\frac{X^TX}{n} + \lambda I)^{-1}\frac{X^Ty}{n}$, and subsequently evaluate objective value trajectories and distance to the optimum.

3.4. Comparison Method

To compare optimization methods fairly, we consider both iteration-based comparisons (GD iterations vs. SGD epochs) and work-normalized comparisons, where computational effort is measured in terms of effective data passes. For example, one GD iteration corresponds to one full pass over the dataset, while one SGD update corresponds to batch size over n data passes. This normalization ensures that differences in convergence are not artifacts of unequal computational budgets. To assess external validity, we also repeat the OLS and ridge experiments on a real-world dataset, namely the UCI Energy Efficiency dataset. This step evaluates whether the trends observed in controlled synthetic settings, particularly the effects of conditioning and regularization, persist in real-world data, where feature correlations and noise are not explicitly controlled.

3.5. Theoretical and Empirical Convergence Rates

We then analyze convergence rates by examining the decay of the optimization error $log||w_k - w^*||$ as a function of iterations. For strongly convex quadratic objectives, theory predicts linear convergence given as $||w_k - w^*|| \leq Cp^k$ where the contraction factor $p\epsilon(0, 1)$ depends on the step size η and the spectrum of the Hessian (H), giving us $p = max\{|1 - \eta\mu|, |1 - \eta L|\}$. The μ and L values represent the smallest and largest values of the Hessian respectively. This result follows from the error recursion $w_{k+1} - w^* = (I - \eta H)(w_k - w^*)$, where the contraction factor is determined by the spectral radius of $I - \eta H$, yielding us p (Hardt & Recht, 2025, p. 78). We estimate empirical convergence rates by fitting linear models to the log-distance curves and compare these slopes to theoretical predictions.

4. Results and Discussion

4.1. Effect of Conditioning and Noise on Gradient Descent

The first set of experiments examines the convergence behavior of full-batch Gradient Descent on the OLS objective under varying conditioning and noise levels. As shown in Figure 1, the objective value decreases monotonically across all settings, but the rate of convergence is strongly dependent on the condition number of the design matrix. For well-conditioned problems, the objective exhibits a steep initial decline, indicating fast linear convergence. As κ increases, convergence becomes progressively slower, with the curves flattening and requiring substantially more iterations to approach the optimum, consistent with the dependence of convergence rates on the spectral properties of $X^T X$. Increasing the noise level raises the minimum achievable objective value, leading to earlier plateaus across all condition numbers. In high-noise regimes, the objective quickly reaches a noise-dominated floor, beyond which further optimization yields negligible improvement. This effect is amplified in ill-conditioned settings, where slow convergence and noise jointly limit the attainable accuracy.

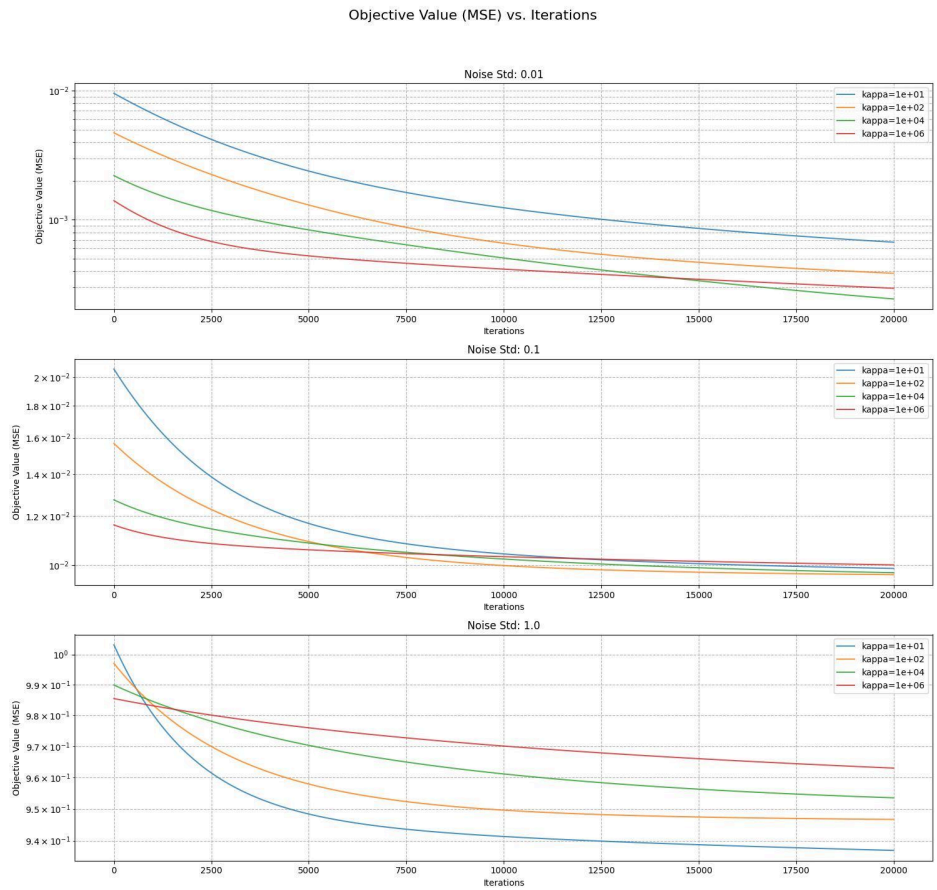

Figure 1: *Objective value versus iterations for standard Gradient Descent under varying condition numbers (kappa) and noise levels. Each subplot corresponds to a different noise standard deviation. Well-conditioned problems converge rapidly, while increasing kappa leads to slower convergence due to ill-conditioning. Higher noise levels raise the error floor, causing earlier plateaus and limiting the achievable accuracy.*

4.2. Effect of Ridge Regularization

To study the effect of regularization on optimization, we consider the ridge regression objective and analyze how varying the regularization parameter λ influences both convergence behavior and solution quality. By introducing an L_2 penalty, ridge regression modifies the spectrum of the Hessian $\frac{1}{n}X^T X + \lambda I$, improving conditioning and, in turn, the convergence properties of Gradient Descent.

Figure 2 shows the evolution of the ridge objective over iterations for different values of λ. As λ increases, the objective becomes better conditioned, leading to faster convergence and earlier stabilization of the iterates. However, this comes at the cost of

a higher final objective value, reflecting the bias introduced by stronger regularization. For small values of λ, the behavior closely resembles that of the unregularized problem, with slower convergence due to ill-conditioning.

Figure 3 complements this analysis by plotting the distance to the ridge optimum. In the absence of regularization, the distance decreases very slowly, indicating that Gradient Descent struggles to make progress in ill-conditioned directions. As λ increases, convergence becomes significantly faster, with the distance to the optimum decaying rapidly, often reaching numerical precision in relatively few iterations. This demonstrates the role of ridge regularization as an implicit preconditioning mechanism that accelerates optimization.

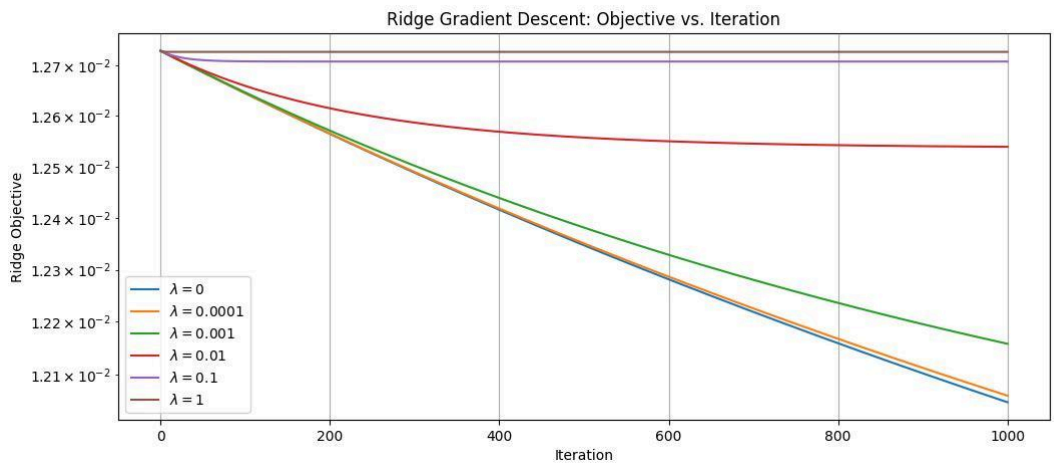

Figure 2: *Ridge objective value versus iterations for different regularization strengths λ. Increasing λ improves conditioning and accelerates convergence, but leads to higher final objective values due to regularization bias.*

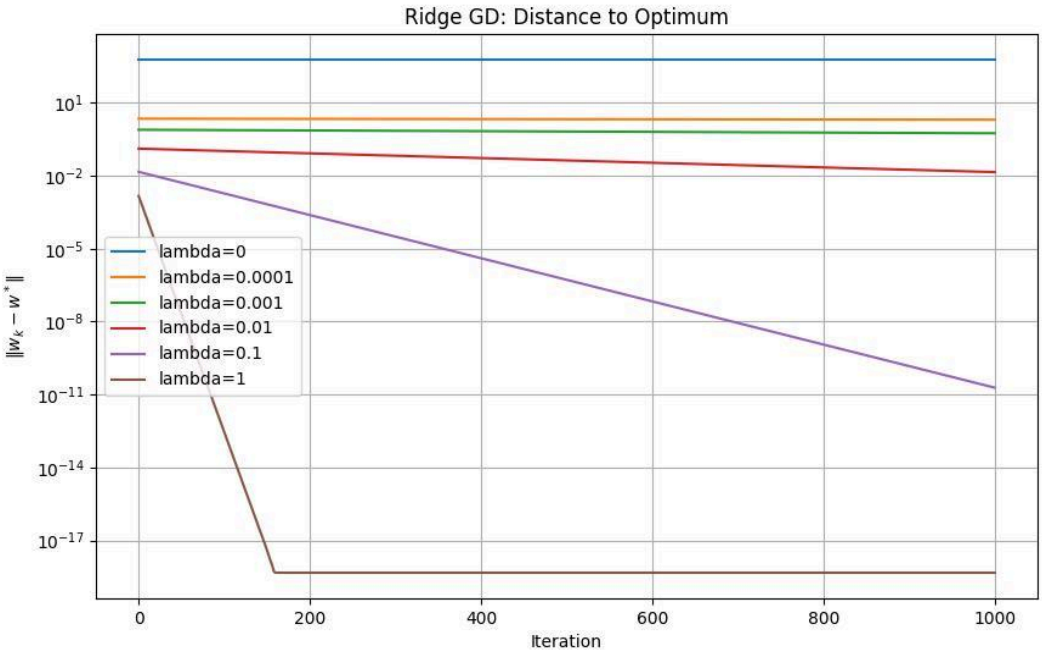

Figure 3: *Distance to the ridge optimum versus iterations. Larger values of λ yield significantly faster convergence, while small or zero regularization results in slow progress due to ill-conditioning.*

4.3. Comparison of GD, Ridge GD, and Ridge SGD

To compare optimization strategies under a common setting, we evaluate full-batch GD, ridge GD, and mini-batch SGD on the same dataset. While ridge GD and ridge SGD optimize the regularized objective, standard GD serves as an unregularized baseline, which allows us to isolate the effect of conditioning and stochastic updates on convergence behavior.

Figure 4 shows the evolution of the objective value for each method. Ridge SGD exhibits a rapid initial decrease in the objective, reflecting the benefit of frequent parameter updates and the ability of stochastic gradients to make fast progress in early iterations. In contrast, ridge GD decreases the objective more steadily, exhibiting the smooth, deterministic convergence expected from full-batch methods. Standard GD shows the slowest decrease, as it operates on the ill-conditioned OLS objective without the stabilizing effect of regularization. As optimization proceeds, the ridge SGD curve

begins to plateau due to the variance inherent in mini-batch updates, while ridge GD continues to decrease smoothly toward the optimum.

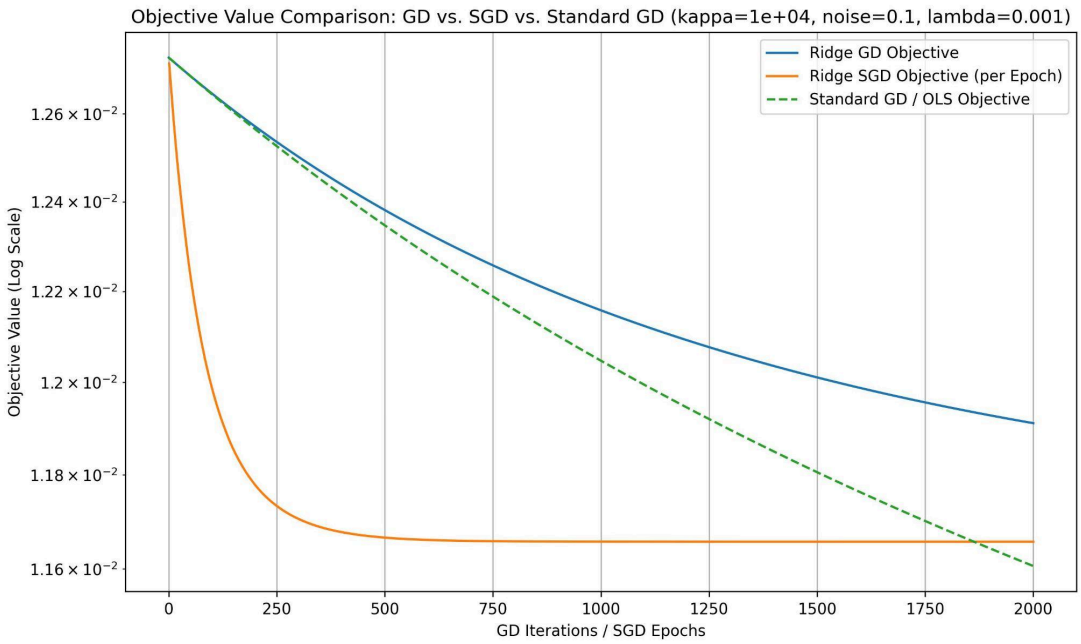

Figure 4: *Objective value versus iterations for standard GD, ridge GD, and ridge SGD. Ridge SGD achieves rapid initial progress due to frequent updates, while ridge GD converges more steadily. Standard GD shows slower convergence due to the absence of regularization.*

Figure 5 presents the distance to the corresponding optimum using a work-normalized metric, where the x-axis represents effective passes over the dataset. This normalization enables a fairer comparison of computational effort between methods. Under this metric, ridge SGD still achieves the fastest initial reduction in distance, indicating that its advantage is not solely due to more frequent updates. However, its progress becomes noisier near the optimum, leading to slower asymptotic convergence. Ridge GD exhibits more stable and consistent convergence, benefiting from improved conditioning due to regularization. Standard GD again performs the worst, with slow decay in distance, highlighting the detrimental impact of ill-conditioning in the absence of regularization.

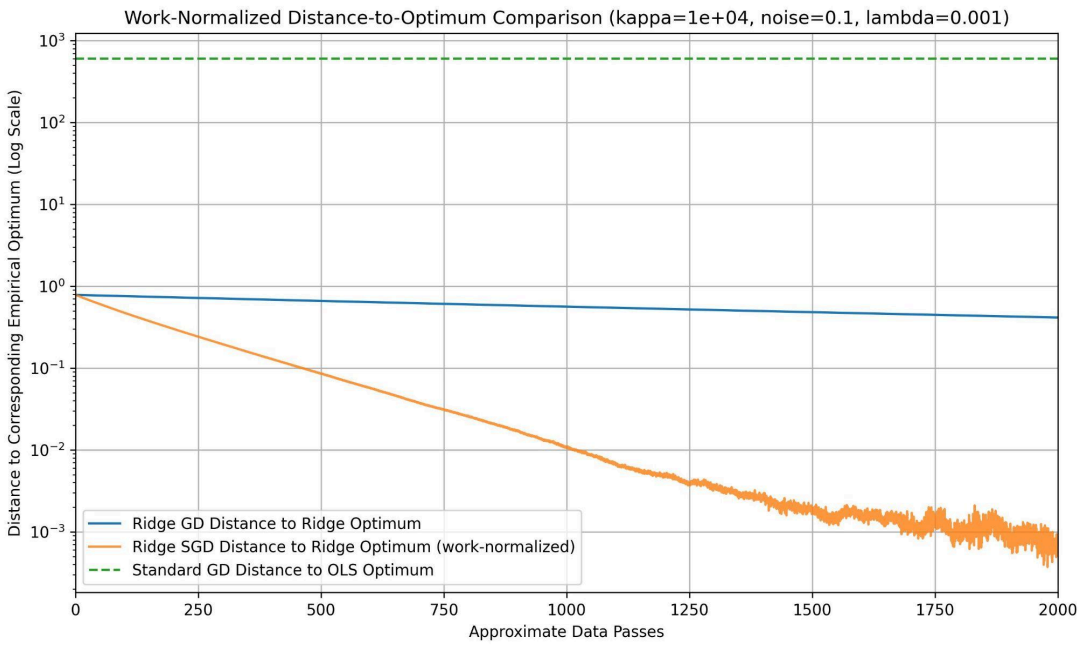

Figure 5: Work-normalized plot showing the *distance to the corresponding optimum versus effective data passes. Ridge SGD achieves the fastest initial reduction even under equalized computational effort, but exhibits higher variability near convergence. Ridge GD shows stable but slower convergence, while standard GD remains slow due to poor conditioning.*

4.4 Theoretical and Empirical Convergence Rates

To further validate the optimization behavior observed in previous experiments, we compare empirical convergence rates with theoretical predictions derived from the spectral properties of the objective. For strongly convex quadratic problems, Gradient Descent is expected to exhibit linear convergence of the for $||w_k - w^*|| \leq Cp^k$.

Figure 6 illustrates this comparison for full-batch methods by plotting $log||w_k - w^*||$ against iterations, along with fitted linear trends representing empirical convergence rates. The ridge-regularized problem exhibits a clear linear decay, with empirical slopes closely matching theoretical predictions, reflecting improved conditioning due to the addition of λ. In contrast, the unregularized (OLS) case shows extremely slow decay, consistent with its poor condition number. While small discrepancies between theoretical and empirical slopes are observed, these can be attributed to practical factors such as finite step sizes and numerical effects. Overall, the results confirm that conditioning controlled via regularization directly governs the convergence rate.

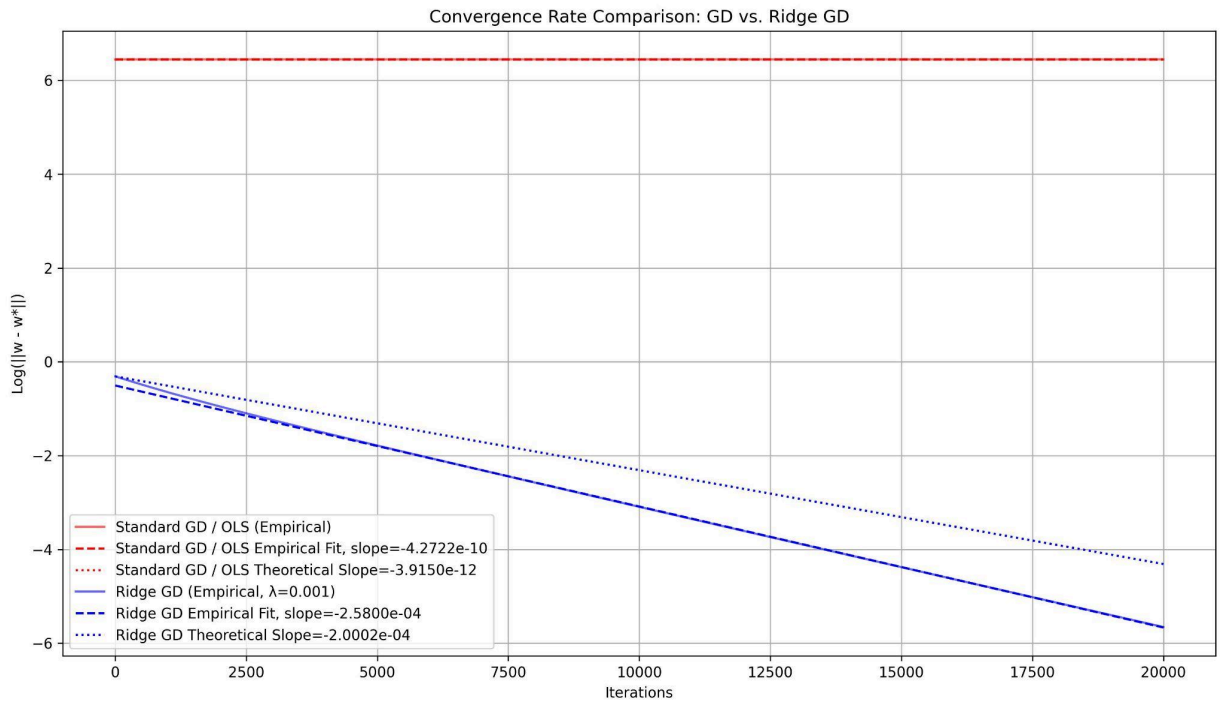

Figure 6: *Log-distance to the optimum versus iterations for standard GD and ridge GD, with fitted linear trends indicating empirical convergence rates. Ridge regularization leads to faster linear convergence consistent with theoretical predictions, while the unregularized case exhibits extremely slow decay due to poor conditioning.*

Figures 7 and 8 extend this analysis to SGD, examining empirical convergence rates at two different granularities, namely for per mini-batch update and per epoch. Figure 7 plots the log-distance to the optimum as a function of individual SGD updates. The empirical curve exhibits an approximately linear decay in the early and intermediate stages, allowing estimation of an effective convergence rate. However, as the iterates approach the optimum, the curve deviates from linearity and flattens due to the inherent variance of stochastic gradients. Figure 8 aggregates SGD progress at the epoch level, where each epoch corresponds to a full pass over the dataset. This representation smooths out the high-frequency noise observed in per-update plots, yielding a clearer view of the overall convergence trend. The resulting curve again shows an initial linear decay, after which progress becomes slower and noisier due to stochastic effects, highlighting the characteristic behavior of SGD.

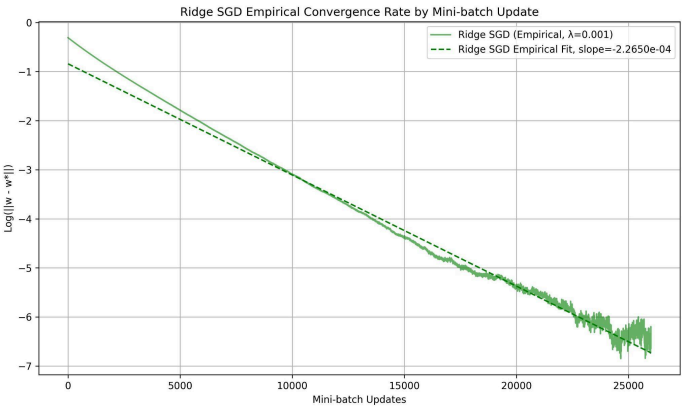

Figure 7: *Log-distance to the ridge optimum versus SGD updates. The curve shows approximately linear decay in early iterations, followed by a plateau due to stochastic gradient variance.*

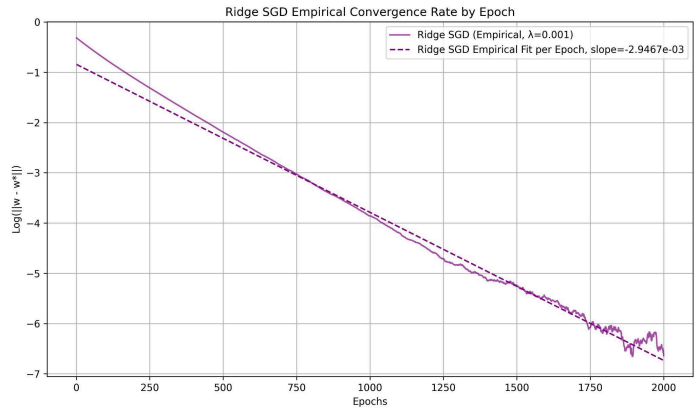

Figure 8: *Log-distance to the ridge optimum versus SGD epochs (full data passes). Aggregating updates at the epoch level smooths stochastic noise, revealing the overall convergence trend of SGD.*

4.5 Real-World Dataset: UCI Energy Efficiency

To assess the extent to which the behaviors observed on synthetic data carry over to real-world settings, we apply the same optimization methods to the UCI Energy Efficiency dataset. This dataset models the heating and cooling load requirements of residential buildings as a function of eight structural features, including relative compactness, surface area, wall area, roof area, overall height, orientation, glazing area, and glazing area distribution (Tsanas & Xifara, 2012). The dataset has already been used in many studies where gradient descent approaches were used to identify the optimal structural configuration of a building to minimize energy consumption (Aklaghi et. al., 2018). In contrast to the synthetic experiments, the conditioning and noise structure of this dataset are not explicitly controlled. However, many of the input features are inherently correlated (e.g., surface area and wall area), which can lead to an ill-conditioned design matrix. This makes the dataset a suitable test case for evaluating how optimization methods behave under realistic feature dependencies. By applying Gradient Descent, ridge Gradient Descent, and stochastic gradient methods to this dataset, we aim to examine whether the effects of conditioning and regularization observed in synthetic settings persist in practice.

Figure 9 shows the evolution of the ridge objective for different values of λ. As in the synthetic case, increasing λ improves the effective conditioning of the problem, resulting in faster convergence and earlier stabilization of the iterates. However, larger values of λ also lead to higher final objective values due to the increased regularization penalty. For small values of λ, the behavior closely resembles that of the unregularized problem, with slower convergence reflecting the underlying conditioning of the dataset.

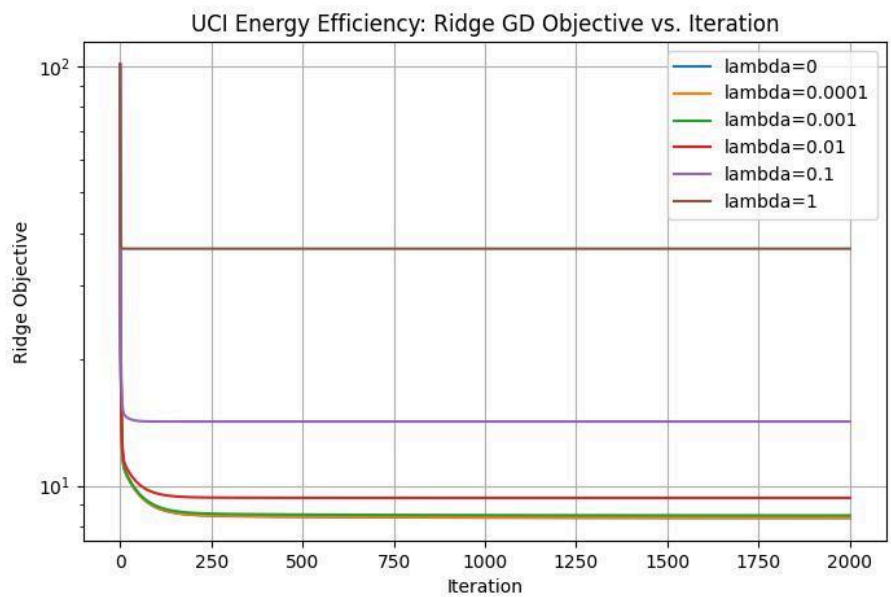

Figure 9: *Ridge objective versus iterations on the UCI Energy Efficiency dataset for different regularization strengths λ. Increasing λ improves conditioning and accelerates convergence, but leads to higher final objective values due to regularization bias.*

Figure 10 compares the distance to the corresponding empirical optimum for standard GD, ridge GD, and ridge SGD. In our implementation, the standard GD case is obtained by evaluating the ridge objective with $\lambda = 0$, ensuring consistency across methods and allowing all distances to be computed within the same optimization framework. Consistent with the synthetic experiments, ridge regularization improves convergence relative to standard GD by effectively modifying the spectrum of the Hessian. Ridge SGD achieves the fastest initial reduction in distance, benefiting from frequent parameter updates and stochastic gradients. When evaluated under a work-normalized metric, this advantage persists, indicating that the improvement is not solely due to

increased update frequency, even though near the optimum, SGD exhibits higher variability and slower asymptotic convergence due to gradient noise, while ridge GD converges more steadily.

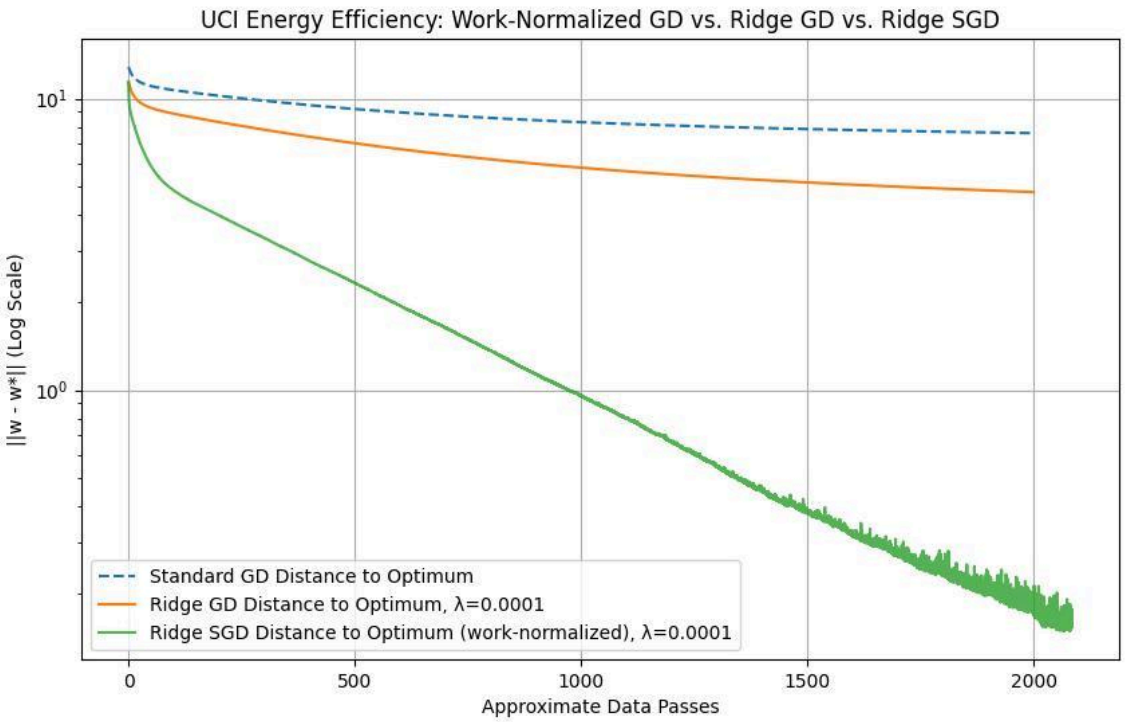

Figure 10: *Work-normalized plot showing the distance to the corresponding empirical optimum versus effective data passes for standard GD (ridge with λ=0), ridge GD, and ridge SGD. Ridge methods converge faster than standard GD, while SGD achieves the fastest initial progress but exhibits higher variability near the optimum.*

5. Future Work

While this study focuses on first-order optimization methods, an important direction for future work is the exploration of second-order and quasi-Newton methods, such as BFGS. These methods are designed to address ill-conditioning by incorporating curvature information into the update rule. In particular, quasi-Newton methods approximate the inverse Hessian matrix iteratively, allowing them to adapt the update direction to the local geometry of the objective function. Unlike standard Gradient Descent, which applies a uniform step size in all directions, quasi-Newton methods effectively rescale the gradient, leading to faster convergence, especially in

ill-conditioned settings. The BFGS algorithm constructs this approximation using only gradient evaluations, avoiding the computational cost of explicitly forming or inverting the Hessian, while still achieving superlinear convergence under standard assumptions. Due to time constraints, BFGS was not incorporated into this study. Implementing these methods would require careful handling of memory and numerical stability, particularly for higher-dimensional datasets. However, if implemented, these methods would provide a useful benchmark for understanding how much of the observed slowdown is due to algorithmic limitations versus intrinsic problem structure.

Additionally, while this work primarily evaluates convergence behavior through objective values and distance to the empirical optimum, future work could extend the analysis to include generalization performance on held-out test data. The current implementation already includes a train/test split for the real-world dataset, but the evaluation focuses on optimization metrics rather than predictive performance. Incorporating test error measurements would allow for a more comprehensive assessment of the trade-off between optimization efficiency and model quality, particularly when comparing regularized and unregularized methods.

6. Conclusion

In this work, we analyzed the optimization behavior of gradient descent and stochastic gradient descent in the context of strongly convex ridge regression. By systematically varying the condition number and noise level, we demonstrated how these factors influence convergence rates and stability. Our results show that poor conditioning significantly slows convergence, while ridge regularization effectively improves optimization performance by enforcing strong convexity. Stochastic gradient descent offers faster initial progress but introduces variability, whereas deterministic gradient descent provides more stable convergence. The consistency between synthetic and real-world experiments highlights the practical relevance of these findings. Overall, this study emphasizes the importance of conditioning, regularization, and algorithm choice in optimization for machine learning.

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